

Which cells are FD2 and FD3? A connectomic assignment of LPT21, Nod3 and LPT42_Nod4

Stage 10 FD2/FD3 Disambiguation

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Summary. Three cell types in the fly connectome are regressive, cholinergic, noduli-family figure cells: LPT42_Nod4, Nod3, and LPT21. Egelhaaf described two regressive figure-detection cells, FD2 and FD3, which share a preferred direction but differ in receptive field and projection side. This report measures the distinguishing properties for all three candidates and assigns each. LPT42_Nod4 is FD3: its receptive field sits laterally with a frontal gap and 90.33% of its output crosses the midline (heterolateral). LPT21 is FD2: its receptive field is frontal, co-located with the FD1 cell, and its output is almost entirely on its own side (1.48% contralateral), the homolateral projection to the ipsilateral posterior optic foci that Egelhaaf reported for FD2. Nod3 is neither cleanly: it is regressive but its output is mixed (44.66% contralateral) and its field is only slightly lateral, so it sits between FD2 and FD3 and is left as an intermediate rather than forced into one identity. The primary numbers come from the offline v783 track; the laterality measurements are confirmed on the live and v630 tracks. Each cell type is a single bilateral pair, so the result rests on left-right agreement and on the thousands of synapses per cell, not on a population average.

1 Introduction

Figure-ground discrimination and the FD cells. A textured object is invisible against a matched background until it moves at a different velocity or phase; relative motion then makes it salient. Reichardt and Poggio showed behaviourally that flies detect and track such figures [4]. Egelhaaf, in a three-part study, described the neurons that could implement this [1, 2, 3]. Part II defined a class of lobula-plate tangential cells, the *figure-detection* (FD) cells, each responding more strongly to a small moving figure than to wide-field motion. Two of them are excited by *regressive* (back-to-front) motion. **FD2** has a frontal excitatory field (peak azimuth 0–10°, reaching the frontal margin of the eye) and is a homolateral cell whose axon projects to the *ipsilateral* posterior optic foci; Egelhaaf notes it is not a “noduli-group” cell and that its large-field organisation could not be resolved [2, p. 201]. **FD3** has a fronto-lateral field (peak 40–50°) that alone among the FD cells spares the most-frontal 10–20° (a frontal gap), and it is a heterolateral “noduli-group” cell whose axon crosses the midline posterior to the noduli to the *contralateral* posterior optic foci [2, p. 202–204]. The two regressive cells therefore share a preferred direction and differ on two axes: the receptive field (frontal versus fronto-lateral with a gap) and the projection side (homolateral/ipsilateral versus heterolateral/contralateral).

The three candidates. FD1 has already been mapped to the cholinergic cell type Nod1, which reads front-to-back motion [6, 5]. Among the remaining noduli-family cells, three are regressive and cholinergic: LPT42_Nod4, Nod3, and LPT21. Because they share direction and transmitter, those properties cannot assign them; the assignment rests on the receptive field and the projection side. This report measures both for all three, together with the small-field selectivity, the morphology, and the whole-family feature match, and reads off each cell’s Egelhaaf identity.

How the assignment is made. Each distinguishing property is measured directly in the connectome: the projection side from where a cell’s output synapses land, the receptive field from the retinotopic positions of its motion inputs relative to the frontal FD1 cell, and the axon path from the cell’s reconstructed shape. We measure these for all three candidates, on both cells of each bilateral pair, on the offline (primary), live, and v630 tracks. The receptive-field comparison is made relative to the FD1=Nod1 anchor in the same coordinate frame, so it needs no conversion into visual angle. Figure 1 fixes the FD2-versus-FD3 distinction; the sections that follow take each property in turn.

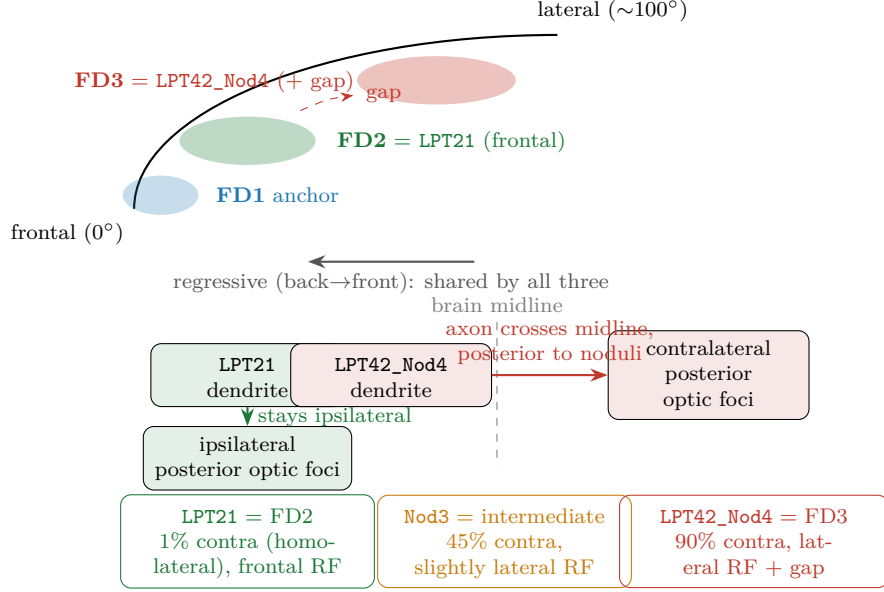


Figure 1: **Three regressive figure-detection candidates, and which Egelhaaf cell each is.** Egelhaaf’s FD2 and FD3 are both excited by regressive (back-to-front) motion, so direction alone cannot tell them apart. They differ in receptive field and projection side: FD2 has a frontal field and a homolateral axon to the ipsilateral posterior optic foci; FD3 has a fronto-lateral field with a frontal gap and a heterolateral axon that crosses the midline posterior to the noduli. The connectome places LPT21 at FD2 (frontal field, essentially all output ipsilateral), LPT42_Nod4 at FD3 (fronto-lateral field with a gap, output contralateral), and Nod3 between them (a mixed projection and a slightly lateral field), so Nod3 is not a clean match to either.

2 The assignment rule

Two Egelhaaf properties separate the regressive figure cells, and each maps onto a connectome measurement. **FD3** requires a fronto-lateral receptive field with a frontal gap (its field sits lateral of the frontal FD1 cell and vacates the frontal band) *and* a heterolateral axon (the large majority of its output crosses the midline). **FD2** requires a frontal receptive field (co-located with the frontal FD1 cell, filling the frontal band, no gap) *and* a homolateral axon (its output stays on its own side, the ipsilateral projection). A cell that is regressive but matches neither pattern cleanly, sitting between the two on both axes, is left as an intermediate rather than forced into an identity.

The rule can conclude differently for each candidate and is not tuned to a desired answer. LPT42_Nod4 is assigned FD3 when it has the lateral field with a gap and the heterolateral axon; LPT21 is assigned FD2 when it has the frontal field and the homolateral axon; and Nod3 is assigned an identity only if it matches one pattern cleanly. The single most important point of method is that output laterality is measured on a continuum rather than a single threshold: a homolateral cell (FD2) keeps its output near zero percent contralateral, a heterolateral cell (FD3) sends most output across, and a cell in between is neither. This is what distinguishes LPT21 (homolateral) from Nod3 (mixed), a distinction a single heterolateral cut-off would miss.

3 Evidence, property by property

We take the properties in turn: first the ones all three candidates share, which cannot assign them, then the two axes that separate FD2 from FD3.

All three cells prefer regressive motion, so direction cannot assign them. A cell’s preferred direction is read from which lobula-plate layer its motion inputs come from. LPT21 draws 99.17% of its motion input from the back-to-front layer, Nod3 97.73%, and LPT42_Nod4 98.58% (Fig. 2). All three are regressive, as both FD2 and FD3 are, so this property is shared and assigns none of them.

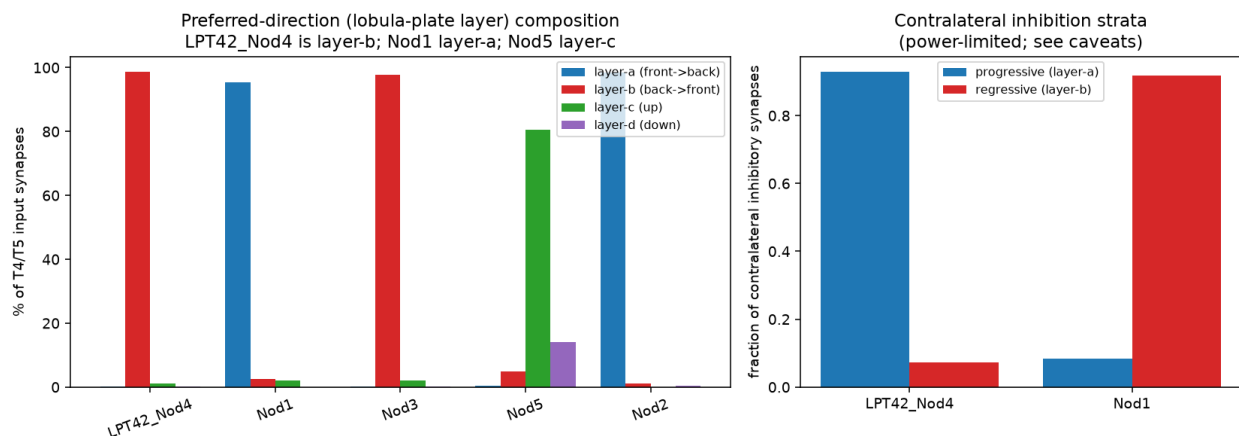


Figure 2: **Preferred direction is shared.** The share of each cell’s motion input from the four directional layers. LPT42_Nod4, Nod3 and Nod1 are shown here (the layer profile of LPT21 is likewise almost entirely back-to-front). Because FD2 and FD3 are both regressive, this panel does not separate the candidates.

All three are cholinergic, so transmitter cannot assign them. Each releases acetylcholine (LPT21 confidence 0.8132, Nod3 0.8784, LPT42_Nod4 0.8943), and each is classified as a visual output neuron. Transmitter and cell class are therefore also shared.

Projection side: LPT21 homolateral, LPT42_Nod4 heterolateral, Nod3 mixed. Egelhaaf’s FD2 is a homolateral cell whose axon projects to the ipsilateral posterior optic foci, whereas FD3 is heterolateral, its axon crossing to the contralateral side. In the connectome LPT21 sends only 1.48% of its output across the midline, the clean homolateral projection of FD2; LPT42_Nod4 sends 90.33%, the heterolateral projection of FD3; and Nod3 sends 44.66%, neither on its own side nor predominantly across, a mixed projection that matches FD2’s homolateral anatomy no better than FD3’s heterolateral one (Fig. 3).

The projection ordering holds on every data track. The fraction of output that is contralateral is sensitive to the data source, so it is reported on more than one track. On the live track LPT21 is 1.58%, Nod3 39.19%, and LPT42_Nod4 80.47%. The exact numbers move between tracks, but the ordering, LPT21 lowest, LPT42_Nod4 highest, Nod3 in between, is preserved.

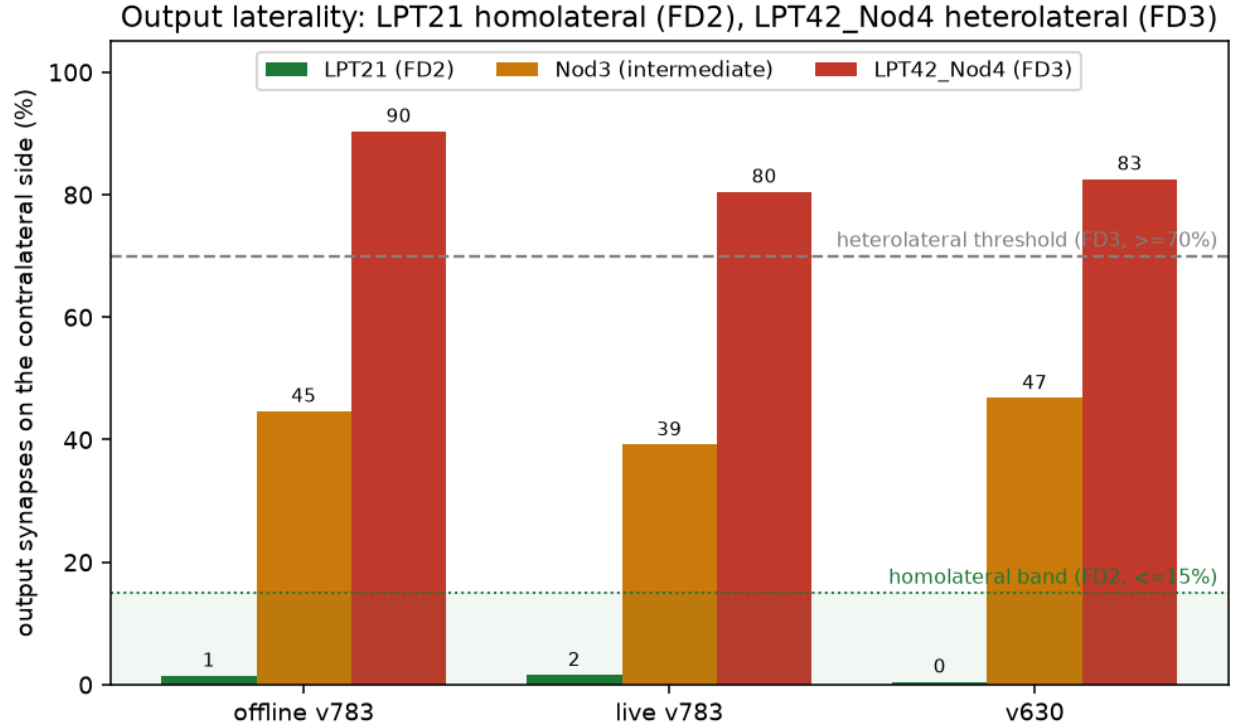


Figure 3: **Output laterality.** The fraction of each candidate’s output synapses on the contralateral side, on the offline, live and v630 tracks. LPT21 lies in the homolateral band (near zero, the FD2 projection); LPT42_Nod4 clears the heterolateral threshold (the FD3 projection); Nod3 sits between them.

Receptive field: LPT21 frontal, LPT42_Nod4 lateral with a gap, Nod3 slightly lateral. Measured against the frontal FD1 cell in the same coordinate frame, the field of LPT21 is co-located with FD1 and fills the frontal band, the frontal field Egelhaaf reported for FD2. The field of LPT42_Nod4 sits far to the side and vacates the frontal band, the fronto-lateral field with a frontal gap that defines FD3. The field of Nod3 is displaced only a few columns from the front and keeps most of its frontal occupancy, so it is neither clearly frontal nor clearly lateral (Fig. 4).

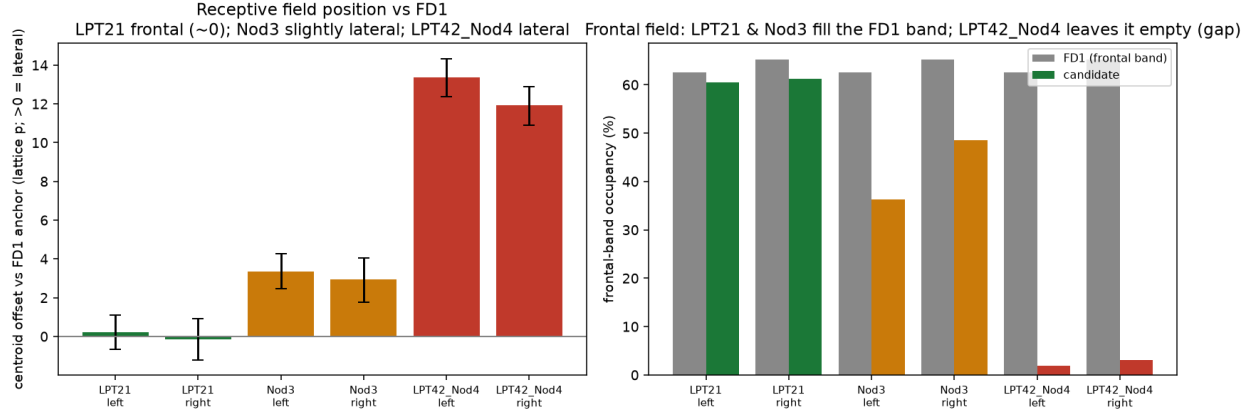


Figure 4: **Receptive field position and the frontal gap.** (left) The centroid of each candidate's input field relative to the frontal FD1 anchor, per side, with 95% confidence intervals: LPT21 near zero (frontal), Nod3 a few columns lateral, LPT42_Nod4 far lateral. (right) Frontal-band occupancy: LPT21 and Nod3 fill the FD1 band, LPT42_Nod4 leaves it nearly empty (the gap).

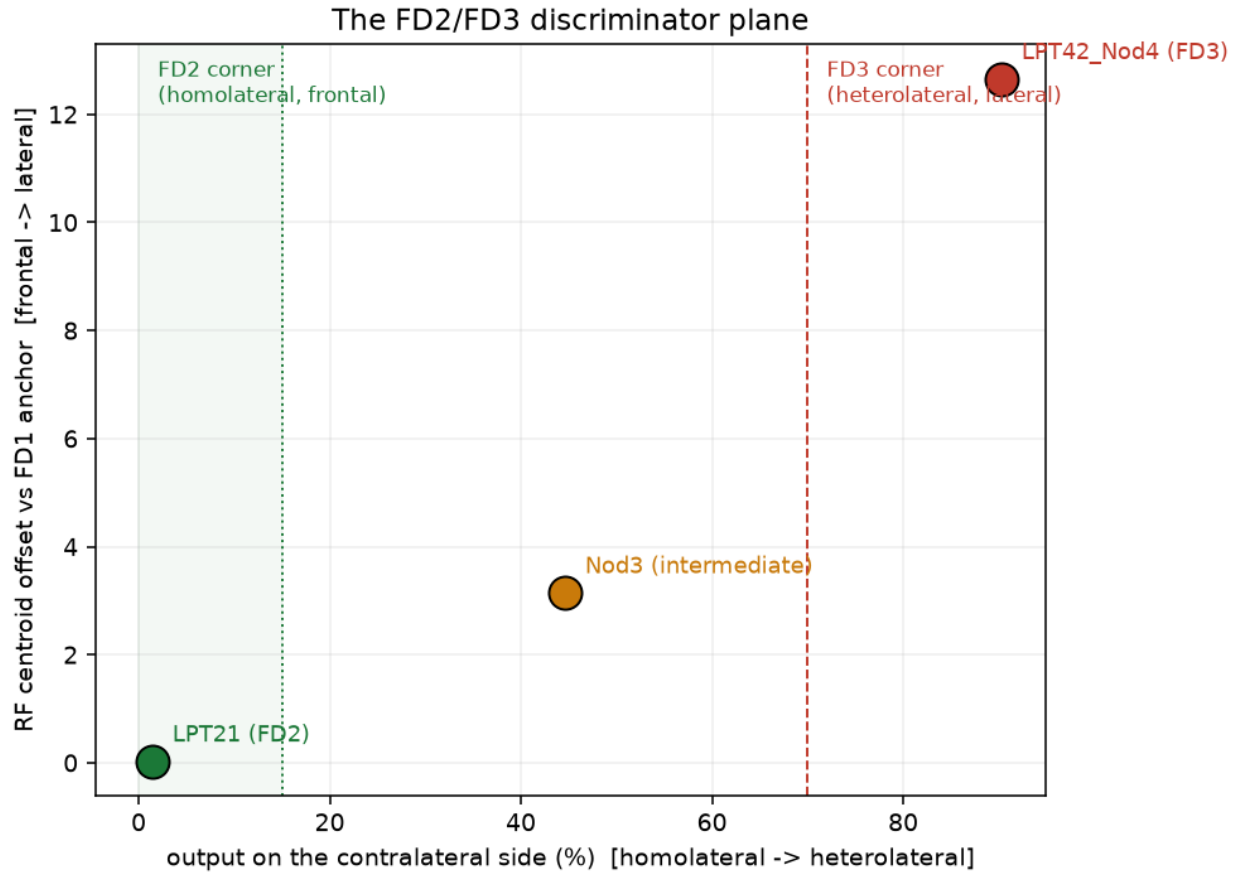


Figure 5: **The FD2/FD3 plane.** Each candidate placed by output laterality (horizontal, homolateral to heterolateral) and receptive-field offset from FD1 (vertical, frontal to lateral). LPT21 sits in the FD2 corner (homolateral, frontal); LPT42_Nod4 in the FD3 corner (heterolateral, lateral); Nod3 between them.

The axon morphology agrees with the projection count. Reconstructing each cell's shape and separating the input (dendritic) from the output (axonal) field shows a medio-lateral shift in every cell, so the shift direction alone does not assign them; what does is how much output lands on the far side. The LPT42_Nod4 axon is displaced across the midline and its output is predominantly contralateral; the LPT21 and Nod3 axons stay on their own side by output count (Fig. 6).

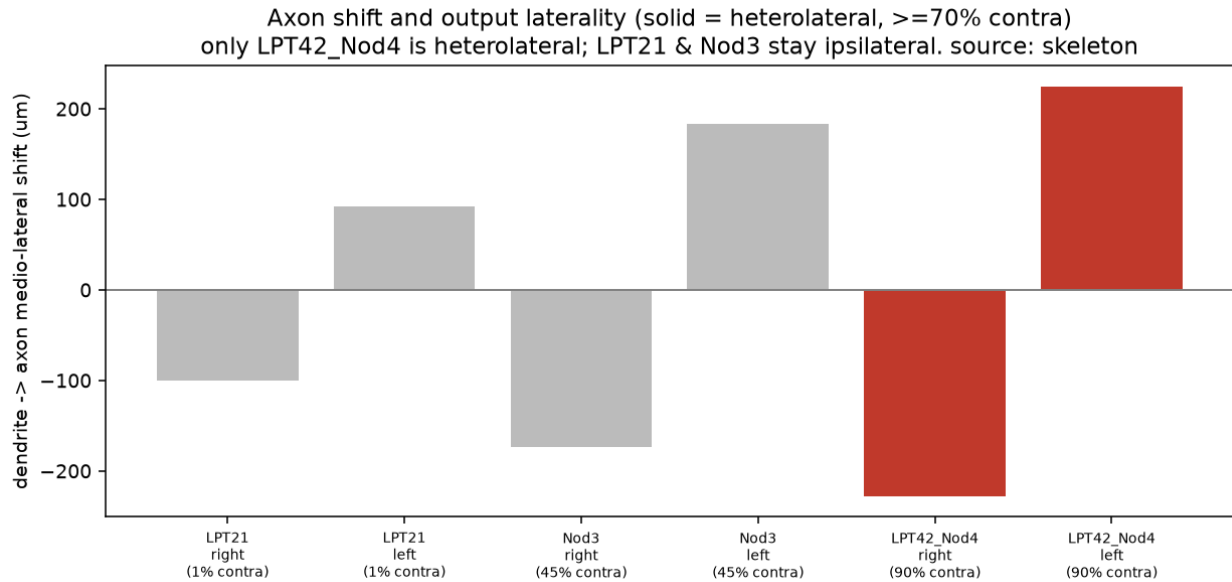


Figure 6: **Axon shift and output laterality.** The dendrite-to-axon medio-lateral shift per cell, with the contralateral output fraction noted beneath. Bars are solid where the cell is heterolateral (at least 70% contralateral); only LPT42_Nod4 is. LPT21 and Nod3 keep their output ipsilateral.

The whole-family feature match, and where it needs refining. Scoring every candidate against all four FD cells on three binary features (direction, the lateral field with a gap, and the crossing axon) makes LPT42_Nod4 the reciprocal-best FD3 (Fig. 7). For FD2 the binary score alone is not decisive: it rates both LPT21 and Nod3 at the FD2 signature, because it collapses every cell below the heterolateral cut-off into one category and so cannot see that LPT21 is cleanly homolateral while Nod3 is mixed. Resolving the tie on the finer measurements, the homolateral projection and the frontal co-location, selects LPT21 as FD2 and leaves Nod3 intermediate. As a check that the feature set can separate the FD cells, the FD1 anchor Nod1 scores best for FD1 (FD1).

FD-family match scores (0-3). FD3 reciprocal best: LPT42_Nod4 (margin 2); FD2 (LPT21/Nod3 tie) broken by homolateral+frontal metrics

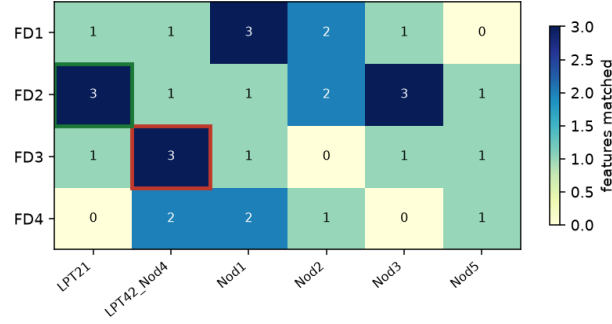


Figure 7: **The family feature match.** Each candidate (columns) scored against each FD cell (rows) by shared binary features (0–3). LPT42_Nod4 (red outline) is the best match for FD3. The FD2 row scores LPT21 (green outline) and Nod3 equally on the binary features; the tie is broken toward LPT21 by the finer homolateral and frontal-co-location measurements.

Three regressive candidates vs Egelhaaf's FD cells				
Property	Egelhaaf reference	LPT21 (FD2)	Nod3 (intermediate)	LPT42_Nod4 (FD3)
Preferred direction (lobula-plate layer)	FD2 and FD3 both regressive (layer-b)	layer-b 99.17%	layer-b 97.73%	layer-b 98.58%
Neurotransmitter	excitatory figure-detection output	acetylcholine (0.8132)	acetylcholine (0.8784)	acetylcholine (0.8943)
Small-field selectivity (bounded vs null)	stronger to small figure than wide field	bounded	bounded	bounded
Receptive-field position vs FD1 anchor	FD2 frontal; FD3 fronto-lateral with a gap	frontal, co-located with FD1 (offset +0.0)	slightly lateral of FD1, no gap (offset +3.1)	lateral of FD1, with frontal gap (offset +12.7)
Frontal gap (unique to FD3)	FD3 alone spares the most-frontal field	no gap (fills frontal band)	no gap (fills frontal band)	gap present (both sides)
Output laterality (contralateral %), offline	FD2 homolateral (ipsi); FD3 heterolateral (contra)	1.48% (ipsilateral (homolateral))	44.66% (mixed / bilateral)	90.33% (contralateral (heterolateral))
Output laterality (contralateral %), live	FD2 homolateral (ipsi); FD3 heterolateral (contra)	1.58%	39.19%	80.47%
Projection target region	FD2 ipsilateral POF; FD3 contralateral POF	ipsilateral POF	mixed / bilateral	contralateral POF
FD-family assignment	each cell should map to one FD identity	FD2 (homolateral + frontal)	intermediate (not clean FD2 or FD3)	FD3 (lateral + gap + heterolateral)
Assignment: FD3=LPT42_Nod4; FD2=LPT21; Nod3=intermediate				
FD2 tie-break: LPT21 is homolateral (contra 1%) and frontal; Nod3 is mixed (45% contra) and slightly lateral, so LPT21 is the clean FD2 and Nod3 is left intermediate.				

Figure 8: **The three-way assignment, at a glance.** Each property, the Egelhaaf reference, and the measured value for each candidate. The shared properties (top) assign none; the projection and receptive-field rows place LPT21 at FD2, LPT42_Nod4 at FD3, and Nod3 between them.

4 Nod3 is an intermediate regressive cell

The earlier version of this analysis, comparing only Nod3 and LPT42_Nod4, assigned Nod3 to FD2 because it is regressive and not FD3. Adding LPT21 shows that assignment was too quick. On the two axes that define FD2, LPT21 is a clean match and Nod3 is not. Nod3's output is 44.66% contralateral, neither the near-zero of a homolateral FD2 cell nor the high fraction of a heterolateral FD3 cell, and its receptive field is displaced only slightly from the front, between FD2's frontal field and FD3's lateral field. Nod3 is therefore best described as an intermediate regressive noduli-family figure cell: it belongs to the same functional group but does not match a single one of Egelhaaf's described FD cells. Whether it is a distinct cell Egelhaaf did not record, a variant of one he did, or a graded member of the regressive set is a question the wiring alone does not settle, and it is left open here rather than forced.

5 The evidence at a glance

Table 1 collects every property next to the measured value for each of the three candidates. The shared properties assign none; the projection and receptive-field rows place LPT21 at FD2, LPT42_Nod4 at FD3, and Nod3 between them.

Table 1: The three regressive candidates against Egelhaaf's FD reference properties. Direction and transmitter are shared; output laterality and receptive-field position assign LPT21 to FD2 (frontal, homolateral), LPT42_Nod4 to FD3 (lateral with a gap, heterolateral), and leave Nod3 intermediate.

Property	Egelhaaf reference	LPT21 (FD2)	Nod3 (interm.)	LPT42_Nod4 (FD3)
Preferred direction (lobula-plate layer)	FD2 and FD3 both regressive (layer-b)	layer-b 99.17%	layer-b 97.73%	layer-b 98.58%
Neurotransmitter	excitatory figure-detection output	acetylcholine (0.8132)	acetylcholine (0.8784)	acetylcholine (0.8943)
Small-field selectivity (bounded vs null)	stronger to small figure than wide field	bounded	bounded	bounded
Receptive-field position vs FD1 anchor	FD2 frontal; FD3 fronto-lateral with a gap	frontal, co-located with FD1 (offset +0.0)	slightly lateral of FD1, no gap (offset +3.1)	lateral of FD1, with frontal gap (offset +12.7)
Frontal gap (unique to FD3)	FD3 alone spares the most-frontal field	no gap (fills frontal band)	no gap (fills frontal band)	gap present (both sides)
Output laterality (contralateral %), offline	FD2 homolateral (ipsi); FD3 heterolateral (contra)	1.48% (ipsilateral (homolateral))	44.66% (mixed / bilateral)	90.33% (contralateral (heterolateral))
Output laterality (contralateral %), live	FD2 homolateral (ipsi); FD3 heterolateral (contra)	1.58%	39.19%	80.47%
Projection target region	FD2 ipsilateral POF; FD3 contralateral POF	ipsilateral POF	mixed / bilateral	contralateral POF
FD-family assignment	each cell should map to one FD identity	FD2 (homolateral + frontal)	intermediate (not clean FD2 or FD3)	FD3 (lateral + gap + heterolateral)

6 What the data can and cannot settle

Each cell type is one bilateral pair. LPT21, Nod3 and LPT42_Nod4 are each a single left/right pair, which is how many the fly has. The result rests on the two cells of each pair agreeing, on the comparison against the FD1 anchor, and on resampling the thousands of synapses each cell carries, not on a population average. **The contralateral fraction is data-source sensitive.** The exact percentage moves between the offline, live and v630 tracks because unannotated local fragments are excluded from the count (the track-to-track movement for LPT42_Nod4 is about 9.9 percentage points); the ordering of the three candidates is preserved on every track, so the assignment does not depend on any one of them. **FD2's large-field organisation was not measured.** Egelhaaf reports that FD2's large-field input could not be resolved, so FD2's only documented properties are its frontal field and its ipsilateral projection; the FD2 assignment of LPT21 rests on those two, not on a contralateral property. **Absolute visual angle is approximate.** The receptive-field argument is built on the relative comparison to the frontal FD1 cell and does not rely on absolute azimuth. **Nod3's positive identity is left open.** It is regressive and a figure cell, but the data place it between FD2 and FD3 rather than on either, and it is reported as an intermediate rather than assigned.

7 Methods and provenance

Connectome. FlyWire FAFB materialization v783 (`synapses_nt_v1`, no cleft threshold), offline track as the primary source, with the live v783 and v630 tracks confirming the laterality measurements. Neuron annotations (cell type, side, super-class, neurotransmitter and confidence, soma coordinates) are the Schlegel/Codex tables. **Preferred direction** is the synapse-weighted T4/T5 lobula-plate layer composition (layer→direction after Fischbach & Dittrich and Maisak et al. [5, 6]). **Output laterality** is the fraction of a cell's output synapses whose postsynaptic partner lies on the opposite side of the brain, reported per track and read as a continuum (homolateral near zero, heterolateral high). **Receptive field:** each cell's T4/T5 inputs carry hex-lattice (p,q) coordinates; the differential test compares the synapse-weighted centroid and frontal-band occupancy against the FD1=Nod1 anchor, with a 2000-sample within-cell bootstrap, and the axis convention is pinned by a self-test requiring the known-frontal FD1 anchor to land frontal. **Morphology:** input and output synapse positions and reconstructed skeletons give the dendrite and axon fields. **Family match:** a binary feature-match over the Nod/LPT `visual_projection` types against the four FD signatures, with the FD2 tie between LPT21 and Nod3 resolved on the homolateral and frontal-co-location measurements. **Reproducibility:** all randomness is seeded (12345). The report is built from the disambiguation JSON. Primary track: `offline`.

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